AMITY UNIVERSITY

AMITY INSTITUTE OF ENVIRONMENTAL SCIENCES

ORGANISES

GREENOVATORS HACKATHON

on

"WASTE TO WEALTH, SMART & SUSTAINABLE INFRASTRUCTURE & NET ZERO AI"

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Attractive Price
₹ 1,00,000

17th – 19th September 2026

Last Date of Registration: 30th May 2026

GREENOVATORS HACKATHON

ON

"WASTE TO WEALTH, SMART & SUSTAINABLE INFRASTRUCTURE & NET ZERO AI"

17th – 19th September 2026

Proposal: **Greenovators Hackathon 2026**

Theme: **Waste to Wealth**
Smart & Sustainable Infrastructure
Net Zero AI

1. Introduction

The goal of the national Greenovators Hackathon 2026 is to inspire students, young researchers, and entrepreneurs to provide workable, technologically advanced, and environmentally friendly solutions to urgent problems. This hackathon highlights two important issues in keeping with India's commitment to the Sustainable Development Goals (SDGs) of the UN:

- Waste to Wealth: Creative ways to turn waste streams into useful resources, such as recycling, upcycling, and the circular economy.
- Smart and Sustainable Infrastructure: Designing and implementing technologically advanced, climate-resilient, and socially inclusive urban infrastructure systems that promote ecological balance and prepare cities for a sustainable future.
- Net Zero AI: Harnessing the power of Artificial Intelligence (AI), Machine Learning (ML), and Data Analytics to accelerate pathways to net zero carbon emissions through predictive modeling, optimization, and digital sustainability solutions.

SDGs 7 (Affordable & Clean Energy), 9 (Industry, Innovation, and Infrastructure), 11 (Sustainable Cities and Communities), 12 (Responsible Consumption and Production), and 13 (Climate Action) align with the Government of India's goal for a sustainable future.

2. Objectives

- To promote smart and sustainable infrastructure that fosters entrepreneurship and innovation in environmental sustainability; to develop models for the circular economy; and to provide solutions for efficient waste management.
- To advance smart city infrastructure solutions that enhance urban quality of life, disaster resilience, and resource efficiency.
- To leverage AI and data-driven technologies for achieving net zero, enabling efficiency in energy systems, carbon footprint monitoring, and sustainable production models within infrastructure.
- To provide a forum for cooperation where policymakers, entrepreneurs, industry, and academics can communicate about smart and sustainable infrastructure.
- To find and develop creative concepts having the potential to be scaled up and commercialised in smart and sustainable infrastructure.

3. Themes & Problem Statements

A. Waste to Wealth

Problem Focus:

- New technology for upcycling and recycling.
- Digital platforms for value recovery, waste collection, and trash segregation.
- The transformation of municipal, e-, and agricultural waste into goods that are valuable.
- Packaging that is sustainable and substitutes for single-use plastics.

Expected Outcomes:

- A waste-to-resource technology prototype.
- Industry and community-based circular business models.
- Scalable concepts to reduce reliance on landfills.

B. Smart & Sustainable Infrastructure

Problem Focus:

- Development of clean, efficient transportation infrastructure and intelligent mobility systems.
- Design and implementation of climate-resilient, energy-efficient urban infrastructure, including buildings, roads, and public spaces.
- Innovations in urban farming infrastructure and integration of green structures within cityscapes to enhance sustainability.
- Deployment of AI and IoT-enabled systems for real-time monitoring and management of critical urban infrastructure, such as waste systems, water supply networks, and air quality.
- Creation of community-driven platforms and tools to foster sustainable urban infrastructure development and maintenance.

Expected Outcomes:

- Prototypes and models of smart infrastructure that seamlessly integrate green design principles, renewable energy sources, and IoT technologies.
- Development of sustainability dashboards and infrastructure monitoring tools to enable city-level management and decision-making.
- Solutions that contribute to reducing urban carbon emissions, enhancing safety, and promoting social inclusivity through resilient infrastructure.

C. Net Zero AI

Problem Focus:

- AI models for real-time carbon footprint monitoring in industries, supply chains, and cities.
- Predictive analytics for renewable energy integration and demand-side management.
- Digital twins and AI-driven simulations for low-carbon urban planning.
- AI-enabled optimization of industrial processes to minimize emissions.
- Climate-fintech solutions using AI for green investments and carbon trading platforms.

Expected Outcomes:

- AI-powered dashboards and predictive models for achieving net zero.
- Intelligent tools that optimize energy, waste, and water management with measurable reductions.
- Demonstration of scalable AI-driven innovations applicable to industries, urban infrastructure, and communities.

4. Target Participants

- University students and research scholars.
- Early-stage startups and innovators.
- Multidisciplinary teams (engineering, environment, design, management).
- NGOs and community-based organizations.

5. Methodology & Structure

Phase 1 – Call for Applications

- Online registration of teams across India.
- Screening of ideas based on innovation, feasibility, and sustainability impact.

Phase 2 – Hackathon Event (3 Days)

- **Day 1:** Orientation, mentoring, and problem briefing.
- **Day 2:** Design sprint, prototype development, and expert guidance.
- **Day 3:** Final pitching before a jury panel comprising industry experts, academicians, and policymakers.

Phase 3 – Post-Hackathon Incubation

- Shortlisted teams to receive mentoring, seed funding support, and incubation opportunities with industry partners.

6. Partnerships & Collaborations

- **Academic Institutions:** Amity University, IITs/NITs, and other partner universities.
- **Industry Partners:** Waste management companies, smart city consultants, renewable energy firms.
- **Government Bodies:** Ministry of Environment, Ministry of Housing & Urban Affairs, and Smart Cities Mission.
- **NGOs & International Agencies:** UNDP, UNEP, and local NGOs working in sustainability.

7. Expected Outcomes & Impact

- **Innovative Solutions:** At least 30+ scalable prototypes addressing waste, smart cities, and net zero AI.
- **Capacity Building:** Training 250+ students and young professionals in sustainable innovation and digital technologies.
- **Policy Alignment:** Recommendations for integration into waste management, AI governance, and urban planning strategies.
- **Societal Impact:** Cleaner cities, reduced emissions, improved quality of urban life, and a digitally enabled low-carbon transition.

8. Budget Estimate (Indicative)

S. No.	Particulars	Amount
1.	Event Organization & Logistics	₹6,00,000
2.	Prizes & Awards	₹2,00,000
3.	Outreach, Media & Documentation	₹2,00,000
4.	Total Estimated Budget	₹10,00,000

9. Alignment with SDGs

- SDG 11: Sustainable Cities and Communities.
- SDG 12: Responsible Consumption and Production.
- SDG 13: Climate Action.
- SDG 9: Industry, Innovation, and Infrastructure.

An innovative effort to unite innovators, entrepreneurs, and stakeholders in the creation of sustainable urban futures is the Greenovators Hackathon 2026. The event's focus on Waste to Wealth and Smart & Sustainable Infrastructure aims to promote a culture of accountability, creativity, and teamwork in addition to generating innovative solutions.